

TRUNCATED SEQUENTIAL TESTS BASED ON ONE-SAMPLE RANK ORDER STATISTICS

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SUMMARY

Sequential tests of $H_0 : \theta = 0$, versus $H_1 : \theta > 0$, where θ is the median of a symmetric distribution, are considered. These tests are based on a general class of one-sample rank order statistics which includes both the Wilcoxon and Normal scores statistics. The tests have a prescribed maximum sample size N , and employ a wide class of stopping boundaries. Under H_0 , the exact limiting type I error and sample size distribution are derived as $N \rightarrow \infty$. Under H_1 , we again let $N \rightarrow \infty$ and derive expressions for the power and sample size as $\theta_N = \theta/\sqrt{N} \rightarrow 0$.

1. Introduction

We are motivated by Miller's (1967, 1970) approach to the following problem: F_θ is an unknown distribution, symmetric about θ . It is desired to discriminate between $H_0 : \theta = 0$ and $H_1 : \theta > 0$ on the basis of a truncated sequential procedure based on independent observations $X_1, X_2, \dots$, each distributed according to F_θ .

Miller considered two procedures based on the Wilcoxon statistic, employing horizontal and square root-type stopping boundaries. The type I errors of these procedures are determined by constants appearing in the boundaries. Miller found the constants by simulation and later, together with Sen (see Miller and Sen (1972)) showed how approximate values could be obtained by direct calculation in the case of a horizontal boundary. He also sought to approximate the power of the test by the probability that Brownian motion with a (given) positive drift crosses the barrier within unit time, and found a remarkable correspondence with the power values obtained by simulation in the case where F_θ is the double exponential distribution.

In this paper we consider truncated sequential procedures based on a class of one-sample rank order statistics and employing rather general stopping boundaries. Asymptotic results are obtained by letting the truncated point tend to infinity. The results of Robbins and Siegmund (1970) on boundary crossing probabilities for Brownian motion make it possible in some cases to find approximate values of the constants determining the type I error.

2. A Limit Theorem

Let $\{T_n, n \geq 1\}$ be a sequence of statistics such that

$$(2.1) T_n = S_n + Z_n$$

with S_n the n -th partial sum of a sequence of i.i.d.r.v. having mean 0 and variance 1, and $n^{-1/2} Z_n \rightarrow 0$ with probability one. Let also $g(t)$ be a non-negative continuous function on $[0,1]$ satisfying

$$(2.2) (i) \quad t^{-1/2} g(t) \uparrow + \infty \text{ as } t \downarrow 0 \text{ in } (0, a_1), 0 < a_1 \leq 1.$$

$$(ii) \quad \int_{0+}^1 t^{-3/2} g(t) \exp(-g^2(t)/8t) \cdot dt < +\infty$$

Theorem 2.1

Under the above assumptions, as $m \rightarrow \infty$:

$$\Pr \{T_n > m^{1/2} g(\frac{n}{m}), \text{ some } 1 \leq n \leq m\} \rightarrow \Pr \{B(t) > g(t), \text{ some } 0 < t \leq 1\}$$

where $\{B(t), t \geq 0\}$ is the standard Brownian motion.

Proof:

Let $\langle x \rangle$ denote the integer part of x . From (2.1) it follows, upon application of theorem 4.1 of Billingsley (1968) that the process $\{m^{-1/2} T_{\langle mt \rangle} - g(\langle mt \rangle/m), 0 \leq t \leq 1\}$ converges in law to $\{B(t) - g(t), 0 \leq t \leq 1\}$ (in the topology of uniform convergence in the space of functions on $[0,1]$ having at most discontinuities of the first kind). Hence, for any $a > 0$, $\sup \{m^{-1/2} T_{\langle mt \rangle} - g(\langle mt \rangle/m) : a \leq t \leq 1\}$ converges in law to $\sup \{B(t) - g(t) : a \leq t \leq 1\}$, the distribution of which has 0 as a continuity point – see lemma 3 of Robbins and Siegmund (1970). This implies

$$(2.3) \quad \lim_{m \rightarrow \infty} \Pr \{T_n > m^{1/2} g(\frac{n}{m}), \text{ some } am \leq n \leq m\} = \Pr \{B(t) > g(t), \text{ some } a \leq t \leq 1\}.$$

$$\text{Now } \Pr \{T_n > m^{1/2} g(\frac{n}{m}), \text{ some } am \leq n \leq m\}$$

$$\leq \Pr \{T_n > m^{1/2} g(\frac{n}{m}), \text{ some } 1 \leq n \leq m\}$$

$$\leq \Pr \{T_n > m^{1/2} g(\frac{n}{m}), \text{ some } 1 \leq n < am\} + \Pr \{T_n > m^{1/2} g(\frac{n}{m}),$$

$$\text{some } am \leq n \leq m\}$$

and the result will follow from this and (2.3), by letting $m \rightarrow \infty$ and then $a \rightarrow 0$, provided

$$(2.4) \quad \lim_{a \rightarrow 0} \lim_{m \rightarrow \infty} \Pr \{T_n > m^{1/2} g(\frac{n}{m}), \text{ some } 1 \leq n < am\} = 0.$$

We have

$$(2.5) \Pr \{n^{-1/2} Z_n > (\frac{m}{n})^{1/2} g(\frac{n}{m})/2, \text{ some } 1 \leq n < am\}$$

$$\leq \Pr \{n^{-1/2} Z_n > a^{-1/2} g(a)/2, \text{ some } 1 \leq n < am\} \text{ for all } a \leq a_1$$

$$\xrightarrow[m \rightarrow \infty]{m} \Pr \{n^{-1/2} Z_n > a^{-1/2} g(a)/2, \text{ some } n \geq 1\}$$

$$\xrightarrow[0]{a} 0, \text{ from (2.2) (i), since } n^{-1/2} Z_n \rightarrow 0 \text{ with probability one,}$$

and

$$(2.6) \lim_{a \rightarrow 0} \lim_{m \rightarrow \infty} \Pr \{S_n > m^{1/2} g(\frac{n}{m})/2, \text{ some } 1 \leq n < am\} = 0,$$

by the obvious analogue of lemma 5 of Robbins and Siegmund (1970). Thus (2.4) holds.

Remarks:

1. Very often in applications the term S_n in (2.1) has higher moments, say of order 4. In this case it is possible to give a very simple proof of (2.6): By theorem 2 of Breiman (1967) $\{S_n, n \geq 1\}$ imitates normal behaviour to within $n^{1/2}$. Hence, as in the proof of (2.5),

$$\lim_{a \rightarrow 0} \lim_{m \rightarrow \infty} \Pr \{S_n - B(n) > m^{1/2} g(\frac{n}{m})/2, \text{ some } 1 \leq n < ma\} = 0$$

while

$$\Pr \{B(n) > m^{1/2} g(\frac{n}{m})/2, \text{ some } 1 \leq n < ma\}$$

$$\leq \Pr \{B(t) > m^{1/2} g(\frac{t}{m})/2, \text{ some } 0 < t < ma\}$$

$$= \Pr \{B(t) > g(t)/2, \text{ some } 0 < t < a\}$$

$$\xrightarrow[0]{a} 0, \text{ by Kolmogorov's test (See Ito and McKean (1965), p. 34).}$$

This proves (2.6).

2. The proof of (2.3) did not require the assumptions (2.2).
3. Theorem 1 continues to hold if T_n and $B(t)$ are replaced by $|T_n|$ and $|B(t)|$.

3. The Tests and their Asymptotic Properties

Let F be a distribution symmetric about 0 and let P_θ be a probability measure under which the r.v. $X_1, X_2, \dots$ are i.i.d. with common distribution F_θ , obtained by translating F through θ . E_θ will denote expectation under P_θ . Let I denote the indicator function and put

$$R_{ni}(a) = \sum_{j=1}^n I(|X_j - a| \geq |X_i - a|)$$

$$T_n(a) = \sum_{i=1}^n (2I(X_i \geq a) - 1) J_n\left(\frac{R_{ni}(a)}{n+1}\right)$$

$$A^2 = \int_0^1 J^2(u) du$$

where $J(u)$, $J_n(u)$, $0 < u < 1$ are defined as in Sen and Ghosh (1971), section 2, and satisfy the conditions given there.

We wish to decide between $H_0 : \theta = 0$ and $H_1 : \theta > 0$ on the basis of a sequential procedure with prescribed maximum sample size $m < \infty$. Let $g(t)$ satisfy (2.2) and define

$$(3.1) \quad N = \text{first } n \geq 1 \text{ at which } T_n(0) \text{ exceeds } m^{1/2} g\left(\frac{n}{m}\right) \\ = +\infty \text{ if } T_n(0) \text{ never exceeds } m^{1/2} g\left(\frac{n}{m}\right).$$

The procedure is then to

$$(3.2) \quad \text{Accept } \theta > 0 \text{ at stage } n \text{ if } N = n \leq m$$

$$\text{Accept } \theta = 0 \text{ at stage } m \text{ if } N > m.$$

and its sample size is $N_m = \min \{N, m\}$. In the following, P_θ^* denotes a probability measure under which the process $\{B(t), t \geq 0\}$ is Brownian with drift θ and unit infinitesimal variance, and we put

$$(3.3) \quad T = \inf \{t > 0 : B(t) > g(t)\} \\ \tau = \min \{T, 1\}.$$

Theorem 3.1

When $m \rightarrow \infty$ we have

$$(i) \quad P_0 \{\text{accept } H_1\} \rightarrow P_0^* \{B(t) > A^{-1}g(t), \text{ some } 0 < t \leq 1\} = \alpha, \text{ (say)}$$

$$(ii) \quad P_0 \{N_m \leq mx\} \rightarrow P_0^* \{\tau \leq x\}, \quad 0 < x < 1$$

$$(iii) P_0 \{N_m = m\} \rightarrow 1 - \alpha$$

$$(iv) E_0 N_m \sim m E_0^* \tau$$

Proof:

Under the conditions in force here, Sen and Ghosh (1974, p. 550) have shown that a representation like (2.1) holds for $T_n(0)/A$. Hence (i) and (ii) follow immediately from theorem 2.1 above, as does (iii) when we bear in mind that $P_0 \{N_m = m\} = 1 - P_0 \{N < m\}$. Since N_m/m converges in law to τ and $N_m/m \leq 1$, part (iv) follows from the dominated convergence theorem.

Theorem 3.2

Let $D(F) = 2 \int_0^\infty \frac{d}{dx} J(2F(x) - 1) dF(x)$. Then for $\theta > 0$, as $m \rightarrow \infty$:

$$(i) P_{\theta m^{-1/2}} \{ \text{Accept } H_1 \} \rightarrow P_{\theta \cdot D(F) \cdot A^{-1}}^* \{ B(t) > A^{-1} g(t), \text{ some } 0 < t \leq 1 \} = 1 - \beta \quad (\text{say}).$$

$$(ii) P_{\theta m^{-1/2}} \{ N_m \leq m x \} \rightarrow P_{\theta \cdot D(F) \cdot A^{-1}}^* \{ \tau \leq x \}, 0 < x < 1.$$

$$(iii) P_{\theta m^{-1/2}} \{ N_m = m \} \rightarrow \beta.$$

$$(iv) E_{\theta m^{-1/2}} N_m \sim m E_{\theta \cdot D(F) \cdot A^{-1}}^* \tau$$

Proof:

Let $V_n(a) = T_n(a) - T_n(0) + \text{an } D(F)$, $V_n = \sup \{ |V_n(a)| : |a| < n^{-1/2} (\log n)^k \}$ with k so large that $\theta \leq (\log n)^k$, $n = 2, 3, 4, \dots$. Then for fixed m and $n = 2, 3, \dots, m$ we have $n^{-1/2} |V_n(-\theta m^{-1/2})| \leq n^{-1/2} |V_n|$, while $n^{-1/2} |V_n| \rightarrow 0$ a.s. P_0 by theorem 4.3 of Sen and Ghosh (1971). Similarly as in the proof of (2.5) then, for $0 < x \leq 1$,

$$\lim_{a \rightarrow 0} \lim_{m \rightarrow \infty} P_0 \{ n^{-1/2} |V_n(-\theta m^{-1/2})| > (\frac{m}{n})^{1/2} (g(\frac{n}{m}) - \theta \cdot D(F) \cdot \frac{n}{m}),$$

$$\text{some } 2 \leq n \leq m x \} = 0$$

Since $g(t) - \theta \cdot D(F)t$ also satisfies (2.2) and since $T_n(a) = T_n(0) - \text{an } D(F) + V_n(a)$, we find as $m \rightarrow \infty$;

$$P_{\theta m^{-1/2}} \{ T_n(0) > m^{1/2} g(\frac{n}{m}), \text{ some } 1 \leq n \leq m x \}$$

$$\sim P_{\theta m^{-1/2}} \{ T_n(0) > m^{1/2} g(\frac{n}{m}), \text{ some } 2 \leq n \leq m x \}$$

$$\begin{aligned}
&= P_O \{T_n(-\theta m^{-1/2}) > m^{1/2} g(\frac{n}{m}), \text{ some } 2 \leq n \leq m x \} \\
&= P_O \{T_n(0) + V_n(-\theta m^{1/2}) > m^{1/2} (g(\frac{n}{m}) - \theta \cdot D(F) \cdot \frac{n}{m}), \text{ some } 2 \leq n \leq m x \} \\
&\rightarrow P_O^* \{B(t) > A^{-1} (g(t) - \theta \cdot D(F) \cdot t), \text{ some } 0 < t \leq x \} \\
&= P_{\theta \cdot D(F) \cdot A^{-1}}^* \{B(t) > A^{-1} g(t), \text{ some } 0 < t \leq x \},
\end{aligned}$$

exactly as in the proof of theorem 2.1. The proof of theorem 3.2 is therefore analogous to that of theorem 3.1.

Remark

For any ξ ,

$$\begin{aligned}
E_{\xi}^* \tau &= \int_0^1 s \cdot P_{\xi}^* \{T \in ds\} + P_{\xi}^* \{T > 1\} \\
&= \int_0^1 s \cdot \left[\frac{d}{ds} P_{\xi}^* \{B(t) > g(t), \text{ some } 0 < t \leq s\} \right] \cdot ds \\
&\quad + P_{\xi}^* \{B(t) \leq g(t), \text{ all } 0 < t \leq 1\}
\end{aligned}$$

Thus, in order to calculate $E_{\xi}^* \tau$, we need the exact crossing probabilities for Brownian motion.

4. Some Special Cases

(4.1) Miller's horizontal barrier procedure

Let C_{α} be the upper $\frac{1}{2}\alpha$ -point of the standard normal distribution ($0 < \alpha < 1$). The test procedure of section 3 was investigated by Miller (1970) in the special case $g(t) = C_{\alpha} \cdot A$ and T_n the Wilcoxon signed-rank statistic. The validity of theorem 3.1 above in this case was established by Miller and Sen (1972). Theorem 3.2 above provides further theoretical justification of the method used by Miller to approximate the power of his test. The expected value of τ in this case has recently been calculated by Reynolds (1975, p. 395). As a further result in this direction we prove:

Theorem 4.1

Let $T_n \equiv T_n(0)$ be the general rank order statistic defined in section 3 and denote by N_H the N of (3.1) when $g(t) = C_{\alpha} A$. Then for every $\theta > 0$, as $m \rightarrow \infty$:

$$E_{\theta} \min \{N_H, m\} \sim m^{1/2} C_{\alpha} \cdot A \cdot \zeta(\theta)^{-1},$$

where $\zeta(\theta) = 2 \int_0^1 J(F(x - \theta) - F(-x - \theta)) dF(x - \theta) - \int_0^1 J(u) \cdot du$.

Proof:

From Sen (1970), theorem 2, we have $n^{-1} T_n \rightarrow \zeta(\theta)$ a.s. P_θ . Now

$$N_H^{-1} T_{N_H} > m^{1/2} C_\alpha \cdot A \cdot N_H^{-1}$$

while

$$(N_H - 1)^{-1} T_{N_H - 1} \leq m^{1/2} C_\alpha \cdot A (N_H - 1)^{-1}$$

Hence, using the fact that $N_H \rightarrow \infty$ a.s. as $m \rightarrow \infty$, it follows that a.s. P_θ :

$$N_H \sim m^{1/2} \cdot C_\alpha \cdot A \cdot \zeta(\theta)^{-1} = \lambda, \text{ say.}$$

Now for $k = 1, 2, \dots$

$$\begin{aligned} P_\theta \{N_H > k\lambda\} &\leq P_\theta \{ \langle k\lambda \rangle^{-1} T_{\langle k\lambda \rangle} < \langle k\lambda \rangle^{-1} m^{1/2} \cdot C_\alpha \cdot A \} \\ &\leq P_\theta \{ \langle k\lambda \rangle^{-1} T_{\langle k\lambda \rangle} - \zeta(\theta) < \zeta(\theta)(2k^{-1} - 1) \} \end{aligned}$$

Choose $k = k_0$ so large that $\zeta(\theta)(2k^{-1} - 1) < -\epsilon$, where $\epsilon > 0$ is arbitrary but fixed. It then follows from Sen (1970), theorem 2, that we can find an m_0 (independent of k_0) such that for all $m \geq m_0$ and $k \geq k_0$:

$$P_\theta \{ \langle k\lambda \rangle^{-1} T_{\langle k\lambda \rangle} - \zeta(\theta) < -\epsilon \} \leq (\rho(\epsilon))^{k\lambda}, 0 < \rho(\epsilon) < 1.$$

Hence

$$\sum_{k=k_0}^{\infty} \sup \{ P_\theta \{ N_H > k\lambda \} : m \geq m_0 \} \leq \sum_{k=k_0}^{\infty} (\rho(\epsilon))^{k \cdot \lambda_0} < +\infty$$

where $\lambda_0 = m_0^{1/2} \cdot C_\alpha \cdot A \cdot \zeta(\theta)^{-1}$. From Bickel and Yahav (1968), lemma 3.2, we conclude that $E N_H \sim \lambda$. The proof is completed by noting that $\min \{ N_H, m \} / N_H \rightarrow 1$ in P_θ probability which follows from the fact that $N_H = o(m)$ a.s. P_θ .

(4.2) A logarithmic barrier procedure

Let $h^{-1}(x)$ denote the function inverse to $h(x) = x^2 + 2 \log \Phi(x)$ and set $g(t) = t^{1/2} h^{-1}(h(z_\alpha) - \log t) A$ where z_α satisfies the equation

$$1 - \Phi(z_\alpha) + \phi(z_\alpha)(z_\alpha + \phi(z_\alpha)/\Phi(z_\alpha)) = \alpha, 0 < \alpha < 1.$$

[$g(t)$ is called a logarithmic barrier since $g(t) \sim (-t \log t)^{1/2} \cdot A$ as $t \rightarrow 0$.] It then follows from Robbins and Siegmund (1970), p. 1413 that the corresponding truncated test has asymptotic type I error α . Also from theorem 3.2 above and theorem 1(ii) of Robbins and Siegmund (1970), after some calculation:

$$\lim_{m \rightarrow \infty} P_{\theta m^{-1/2}} \{\text{accept } H_1\} = 1 - \Phi(z_\alpha - b) + b^{-1} \phi(z_\alpha - b) \\ - b^{-1} \phi(z_\alpha) \Phi(z_\alpha - b) \Phi(z_\alpha)^{-1}$$

where $b = \theta \cdot D(F) \cdot A^{-1}$. It seems impossible to express h^{-1} analytically in some nice form, and hence explicit expressions for $E^* \tau$ will be hard to come by. However, proceeding similarly as in theorem 4.1, it can be shown that (N_L denoting the N of (3.1) in this case) $E_\theta \min(N_L, m) \sim \log m \cdot \zeta(\theta)^{-2}$, $\theta > 0$.

(4.3) Square root barrier

Miller (1967) has considered a sequential Wilcoxon test based on a square root type barrier. A slight reformulation of his procedure allows us to deduce some asymptotic theory. However, the relevant crossing probabilities for the Broanian motion are not explicitly known, making comparison with Miller's simulation-results difficult.

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